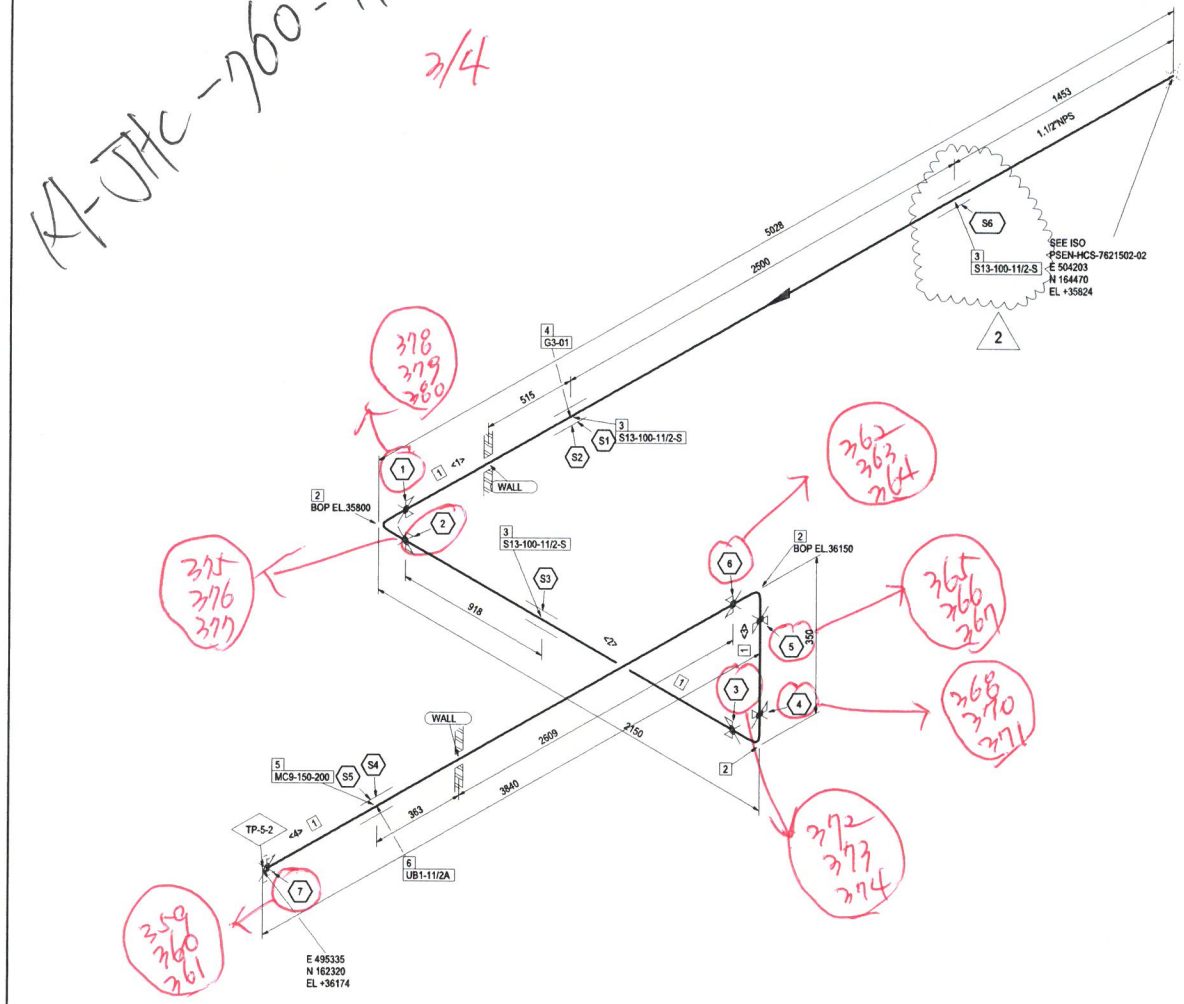
K1-UHC-960-HCS1-006

3/4



NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°C)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°C)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
1.5	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
2	26-09-25	ISSUED FOR REVISION	G.Y.LEE	M.H.SUNG	T.H.LIM
1	22-09-25	ISSUED FOR REVISION	G.Y.LEE	M.H.SUNG	T.H.LIM
0		ISSUED FOR INFORMATION	G.Y.LEE	M.H.SUNG	T.H.LIM

NOTE : THE ORIGINAL AND ALL COPIES OF THIS DRAWING TOGETHER WITH COPYRIGHT ARE THE SOLE PROPERTY OF LEC, SEOUL, KOREA AND SHALL NOT BE USED FOR ANY PURPOSE WITHOUT WRITTEN PERMISSION OF LEC.

BILL OF MATERIALS

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE SMLS ASTM A312-316/316L SCH40S PE	1.1/2"	11.2 M	S98	P	PA02-000000958
2	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	1.1/2"	3	S98	E	PB99-000038443
3	S13-100-11/2-S	1.1/2"	3			S13
4	G3-01	1.1/2"	1			G3
5	MC9-150-200	1.1/2"	1			MC9
6	UB1-11/2A	1.1/2"	1			UB1-A

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	4996	1.1/2		<2>	2087	1.1/2	
<3>	287	1.1/2		<4>	3801	1.1/2	

F3078

Project PROVIDENCE

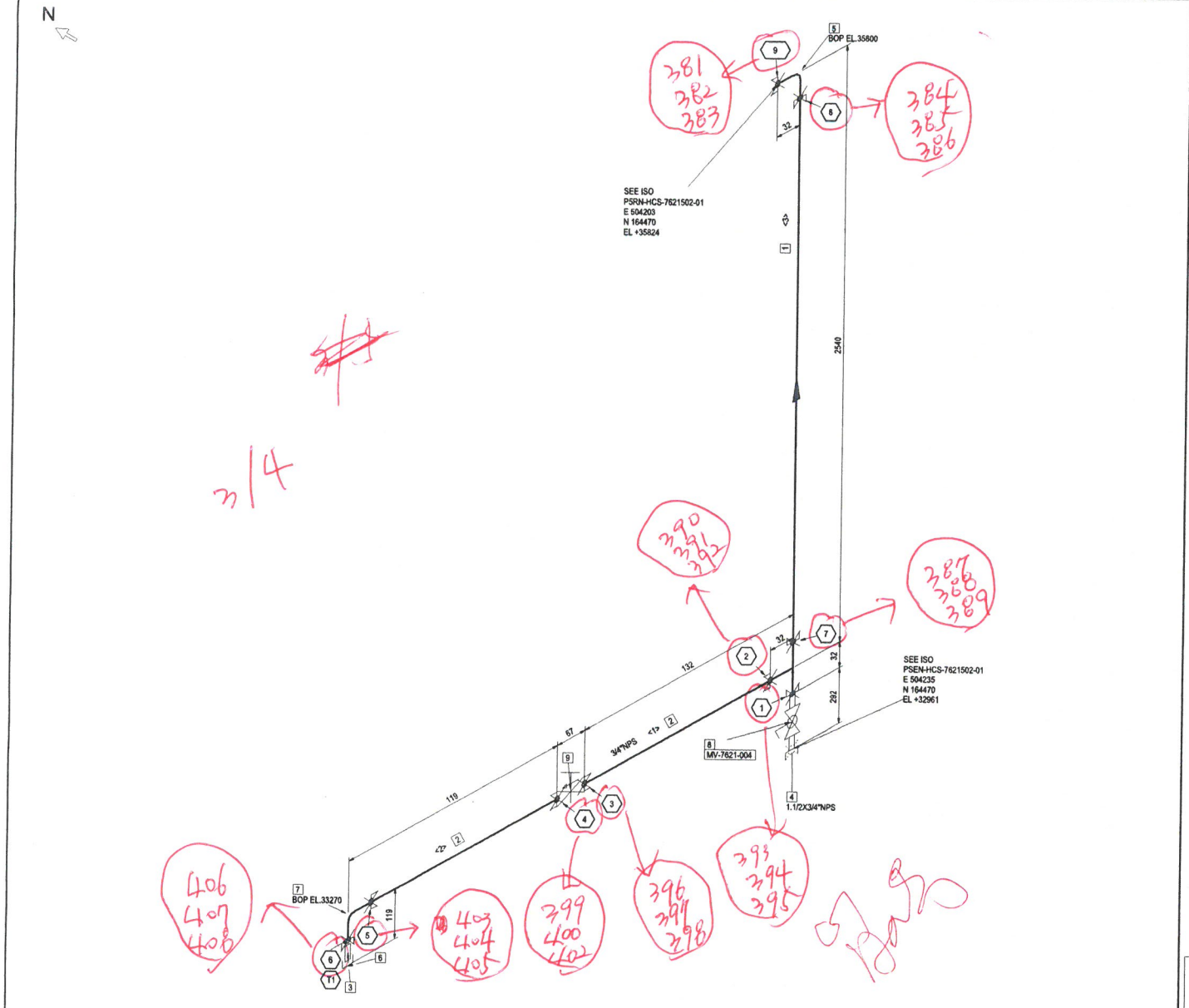
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-09-26

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P5RN
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480	FLUID	HCS
GAD NO.	-			LINE NO.	1.5-HCS-7621502-S98-IHI		
DWG NO.	P5RN-HCS-7621502-01			SH'T NO.	1 OF 1	REV.	2

PIPE modified at 15:16:10 Sep 2025, Drawing Created at 15:15:04 10 September 2025
PIPE NAME : 1.5-HCS-7621502-S98-IHI_P5RN-01



BILL OF MATERIALS

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE SMLS ASTM A312-316/316L SCH40S PE	1.1/2"	2.5 M	S98	P	PA02-000000958
2	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	0.2 M	S98	P	PA02-000000956
3	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
4	REDUCING TEE SW CL3000 ASTM A182-F316/316L	1.1/2X3/4"	1	S98	TR	PB99-000038852
5	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	1.1/2"	1	S98	E	PB99-000038443
6	CAP SCRD (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
7	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	1	S98	E	PB99-000038441
8	BALL VALVE SW ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLO ATING BALL FB 3 PIECE BODY LV. OP. (VAB060T)	1.1/2"	1	S98	VL	PP04-000007762
9	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	100	3/4		<2>	100	3/4	
<3>	2476	1.1/2					

F1027

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-09-26

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (℃)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (℃)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
1.5	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
2	26-09-25	ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
1		ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
0	15-07-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

PROJECT: PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480
CAD NO.	-			LINE NO.	1.5-HCS-7621502-S98-IHI
DWG NO.	PSEN-HCS-7621502-02			SHT NO.	1 OF 1

CWA NO. PSEN
FLUID HCS
REV. 2

PIPE modified at 13:47:25 Sep 2025 , Drawing Created at 13:47:46 25 September 2025
PIPE NAME : 1.5-HCS-7621502-S98-IHI_PSEN-02

NOTE : THE ORIGINAL AND ALL COPIES OF THIS DRAWING TOGETHER WITH COPYRIGHT ARE THE SOLE PROPERTY OF LEC, SEOUL, KOREA AND SHALL NOT BE USED FOR ANY PURPOSE WITHOUT WRITTEN PERMISSION OF LEC.